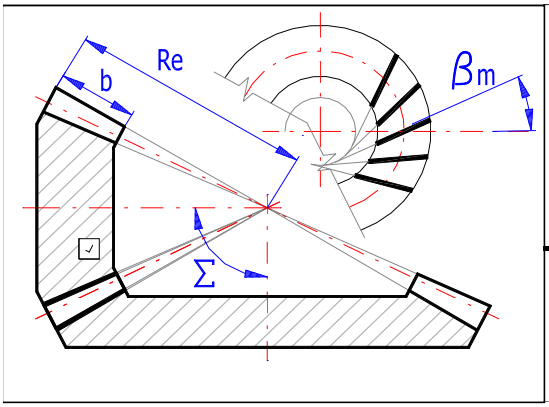
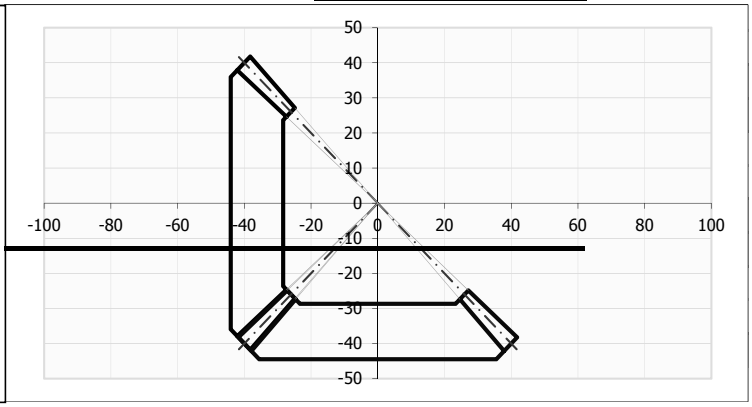
Bevel gearing with straight, oblique and curved teeth [mm/DIN]

			Pinion	Gear	
i	**Calculation without errors.**				
ii	☐ **Project information**				
?	**Input section**				
1.0	☑ **Options of basic input parameters**				
1.1	Transferred power	Pw [kW]	0.066	0.065	
1.2	Speed (Pinion / Gear)	n [/min]	90.0	90.0	[/min]
1.3	Torsional moment (Pinion / Gear)	Mk [Nm]	7.00	6.94	[Nm]
1.4	Transmission ratio / from table	i	1.00		
1.5	Actual transmission ratio / deviation	i	1.0000	0.00%	
2.0	☑ **Options of material, loading conditions, operational and production parameters**				
2.0	Material identification according standard :		DIN		
2.1	Material of the pinion : B...Grey cast iron GG-30 (Rm=300 MPa)				
2.2	Material of the gear : B...Grey cast iron GG-30 (Rm=300 MPa)				
2.3	Loading of the gearbox, driving machine - examples		B...Light shocks		
2.4	Loading of gearbox, driven machine - examples		B...Light shocks		
2.5	Type of gearing mounting		A...Overhung both wheels - type 1		
2.6	Accuracy grade - DIN3965 / ISO1328 |Ra max|v max		3 / 4.....(Ra max.= 0.4 / v max.= 5)		
2.7	Coefficient of one-off overloading	KAS	2.00		
2.8	Desired service life	Lh	20000		[h]
2.9	Coefficient of safety (contact/bend)	SH / SF	1.30	1.60	
2.10	Automatic design		A. Spur gearing		
3.0	☑ **Parameters of the tooth profile, gearing type**				
3.1	Guiding curve of the toothing (Type of toothing)		A,B. Straight line, type I (Standard straight and oblique toothing)		☑
3.2	Addendum - Coefficient of the height of the tooth head	ha*	1.000	1.000	[modul]
3.3	Unit head clearance	c*	0.200	0.200	[modul]
3.4	Recommended coefficient of the root radius		0.304	0.304	[modul]
3.5	Coefficient of the root radius	rf*	0.304	0.304	☑ [modul]
4.0	☑ **Design of a module (Diametral Pitch) and geometry of toothing**				
4.1	Number of teeth Pinion / Gear	z	32	32	
4.2	Angle of shaft axes	Σ	90	90	[°]
4.3	A. Transverse pressure angle	α	20.0	20.0	[°]
4.4	Base helix angle	βm	0.0	0	[°]
4.5	Direction of the teeth pitch (pinion)		Right-Hand		
4.6	Width of toothing to the surface straight line of the cone (b/Re)				
4.7	Width of toothing to the surface straight line of the cone (b/Re)	Re/b	0.35	< 0.35	
4.8	A. Transverse module (outer)	met	2.500		[mm]
4.9	Face width / max. recommendet value	b	19.8	< 19.8	☑ [mm]
4.10	Approximate weight of the gearing	m	0.848		[kg]
4.11	Minimum coefficient of safety	SH / SF	1.329	2.472	



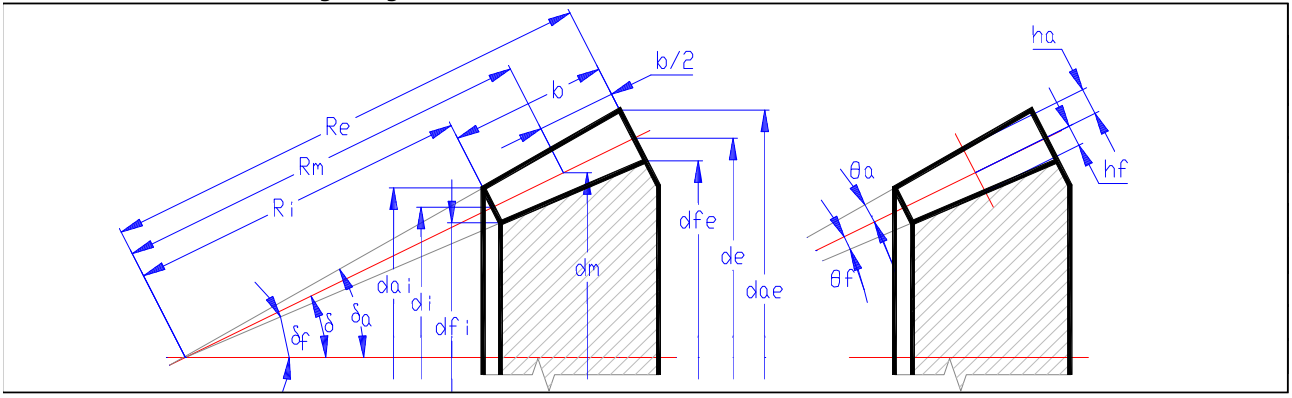


5.0	**Correction of toothing (Addendum modification)**				
5.1	Correction type				
5.2	Recommendet value	x1 / xτ1	0.066	0.000	
5.3	- Permissible undercutting of teeth (min. value)	x1 / x2	-1.8136	-1.8136	

5.4	- Preventing undercutting of teeth (min. value)	x1 / x2	-1.6469	-1.6469
5.5	Pinion addendum modification coefficient setting			
5.6	Addendum modification coefficient Pinion / Gear	x1 / x2	0.0000	0.0000
5.7	Tooth thicknes modification coefficient	$x\tau_1$ / $x\tau_2$	0.0000	0.0000
5.8	Total contact ratio	$\varepsilon\gamma$	1.7369	
5.9	Unit tooth thickness on the tip diameter	sae*	1.0243	1.0243
5.10	Safety coefficient for surface durability	SH	1.329	1.329
5.11	Safety coefficient for bending durability	SF	2.47	2.47

Results section

6.0 ☑ Basic dimensions of gearing



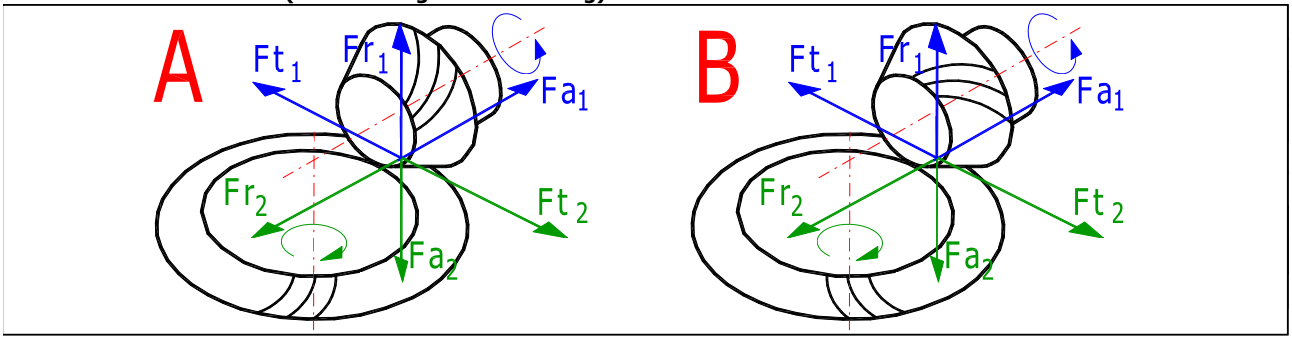
6.1	Number of teeth Pinion / Gear		z	32	32	
6.2	Transverse module (outer, midle, inner)	met,mmt,mit	2.5000	2.0625	1.6250	[mm]
6.3	Normal module (outer, midle, inner)	men,mmn,min	2.5000	2.0625	1.6250	[mm]
6.4	Cone length (outer, midle, inner)	Re,Rm,Ri	56.569	46.669	36.769	[mm]
6.5	Pitch cone angle		δ	45.0000	45.0000	[°]
6.6	Addendum cone angle		δa	47.5305	47.5305	[°]
6.7	Dedendum cone angle		δf	41.9643	41.9643	[°]
6.8	Tip diameter (outer)		dae	83.536	83.536	[mm]
6.9	Tip diameter (midle)		dam	68.916	68.916	[mm]
6.10	Tip diameter (inner)		dai	54.297	54.297	[mm]
6.11	Pitch diameter (outer)		de	80.000	80.000	[mm]
6.12	Pitch diameter (midle)		dm	65.999	65.999	[mm]
6.13	Pitch diameter (inner)		di	51.999	51.999	[mm]
6.14	Root diameter (outer)		dfe	75.757	75.757	[mm]
6.15	Root diameter (midle)		dfm	62.499	62.499	[mm]
6.16	Root diameter (inner)		dfi	49.241	49.241	[mm]
6.17	Addendum angle		θa	2.5305	2.5305	[°]
6.18	Dedendum angle		θf	3.0357	3.0357	[°]
6.19	Addendum (outer)		hae	2.5000	2.5000	[mm]
6.20	Addendum (midle)		ha	2.0625	2.0625	[mm]
6.21	Addendum (inner)		hai	1.6250	1.6250	[mm]
6.22	Dedendum (outer)		hfe	3.0000	3.0000	[mm]
6.23	Dedendum (midle)		hf	2.4750	2.4750	[mm]
6.24	Dedendum (inner)		hfi	1.9499	1.9499	[mm]
6.25	Normal pressure angle		αn	20.0000		[°]
6.26	Transverse pressure angle		αt	20.0000		[°]
6.27	Helix angle		β	0.00		[°]
6.28	Base helix angle		βb	0.0000		[°]
6.29	Pressure angle at the pitch cylinder		αwn	20.0000		[°]
6.30	Transverse pressure angle at the pitch cylinder		αwt	20.0000		[°]
6.31	Circular pitch		pe	7.854		[mm]
6.32	Transverse circular pitch		pte	7.854		[mm]
6.33	Tooth thickness on the pitch diameter		sne	3.9270	3.9270	[mm]
6.34	Tooth thickness on the pitch diameter		sn	3.2397	3.2397	[mm]
6.35	Tooth thickness on the pitch diameter		sni	2.5525	2.5525	[mm]

6.36	Tooth thickness on the tip diameter	sae	2.5607	2.5607	[mm]	
6.37	Tooth thickness on the tip diameter	sa	2.1126	2.1126	[mm]	
6.38	Tooth thickness on the tip diameter	sai	1.6644	1.6644	[mm]	
6.39	Unit tooth thickness on the tip diameter	sae*	1.0243	1.0243	[modul]	
7.0	☑ **Virtual spur gear toothing**					
7.1	Number of teeth of a virtual wheel with oblique teeth	zvn'	45.255	45.255		
7.2	Number of teeth of a virtual wheel with straight teeth	zv	45.255	45.255		
7.3	Reference diameter	dv'	93.337	93.337	[mm]	
7.4	Tip diameter	dva'	97.462	97.462	[mm]	
7.5	Base diameter	dvb'	87.708	87.708	[mm]	
7.6	Root diameter	dvf'	88.387	88.387	[mm]	
7.7	Virtual center distance	av	93.3371		[mm]	
7.8	Virtual Gear Ratio	iv	1.0000			
8.0	☑ **Qualitative indexes of a gearing**					
8.1	Transverse contact ratio / overlap ratio	$\varepsilon_\alpha \mid \varepsilon_\beta$	1.7369	0.0000		
8.2	Total contact ratio	$\varepsilon\gamma$	1.7369			
8.3	Resonance speed	nE1	9825.09		[/min]	
8.4	Resonance ratio	N	0.01			
8.5	Approximate weight of the gearing	m	0.8480		[kg]	
8.6	Efficiency of the gearing	μ	99.15%			
8.7	Selected / Recomended lubricant viscosity	v50	92	92	☑ [mm2/sec]	
9.0	☑ **Coefficients for safety calculation**					
	Common for the gearing					
9.1	Stiffness of a tooth pair	c'	5.00			
9.2	Meshing stiffness per unit face width	cγ	7.76			
9.4	Application factor	KA	1.350			
9.5	Dynamic factor	KV	1.004			
9.6	Number of cycles	NK	1.08E+08	1.08E+08		
	For pitting safety calculation					
9.7	Face load factor (contact stress)	$K_{H\beta}$	1.595			
9.8	Transverse load factor (contact stress)	$K_{H\alpha}$	1.326			
9.9	Total factor of additional loads	KH	2.866			
9.10	Elasticity factor	ZE	138.50			
9.11	Zone factor	ZH	2.495			
9.12	Helix angle factor	Zbeta	1.000			
9.13	Contact ratio factor	Zeps	0.869			
9.14	Size factor	ZX	1.000	1.000		
9.15	Lubricant factor	ZL	0.989	0.989	Mineral oil	
9.16	Peripheral speed factor	ZV	0.879	0.879		
9.17	Roughness factor affecting surface durability	ZR	1.030	1.030	Automatic	
9.18	Bevel gear factor (flank)	ZK	0.850			
9.19	Life factor for contact stress	ZN	0.926	0.926	n=∞; ZN=0.85	
9.20	Single pair tooth contact factor	ZB / ZD	1.001	1.001		
	For bending safety calculation					
9.21	Face load factor (root stress)	$K_{F\beta}$	1.595			
9.22	Transverse load factor (root stress)	$K_{F\alpha}$	1.467			
9.23	Total factor of additional loads	KF	3.171			
9.24	Helix angle factor	Ybeta	1.000			
9.25	☑ Contact ratio factor	Yeps	0.682		☑	
9.26	Notch sensitivity factor	Ydelta	1.003	1.003		
9.27	Size factor	YX	1.000	1.000		
9.28	Tooth-root surface factor	YR	1.020	1.020		
9.29	Alternating load factor	YA	1.000			
9.30	Production technology factor	YT	1.000			
9.31	Life factor for bending stress	YN	0.850	0.850		
9.32	Tip factor, equal (YFa YSa)	YFS	3.758	3.758		

10.0 ☑ Safety coefficients

10.1	Safety coefficient for surface durability	SH	1.33	1.33	
10.2	Safety coefficient for bending durability	SF	2.47	2.47	
10.3	Safety in contact in one-time overloading	SHst	3.02	3.02	
10.4	Safety in bending in one-time overloading	SFst	3.55	3.55	
10.5	Nominal contact stress	SigmaH0	132.59		
10.6	Contact stress	SigmaH	224.74	224.74	
10.7	Pitting stress limit	SigmaHG	298.63	298.63	
10.8	Permissible contact stress	SigmaHP	229.72	229.72	
10.9	Nominal tooth-root stress	SigmaF0	13.32	13.32	
10.10	Tooth-root stress	SigmaF	42.22	42.22	
10.11	Tooth-root stress limit	SigmaFG	104.38	104.38	
10.12	Permissible bending stress	SigmaFP	65.23	65.23	

11.0 ☑ Force conditions (forces acting on the toothing)



11.1	Tangential force	Ft	212.22		[N]
11.2	Normal force	Fn	225.84		[N]
11.3	Axial force - (rotation acc. to the picture)	Fa	54.62	54.62	[N]
11.4	Radial force (rotation acc. to the picture)	Fr	54.62	54.62	
11.5	Axial force (rotation opposite to the picture)	Fa	54.62	54.62	[N]
11.6	Radial force (rotation opposite to the picture)	Fr	54.62	54.62	
11.7	Peripheral speed on the pitch diameter	v | vmax	0.31	< 5	[m/s]
11.8	Specific load / Unit load	wt | wt*	17.02	8.25	[N/mm | MPa]

12.0 ☑ Parameters of the chosen material

12.1	Density	Ro	7870	7870	[kg/m^3]
12.2	Young's Modulus (Modulus of Elasticity)	E	113	113	[GPa]
12.3	Tensile Strength, Ultimate	Rm	300	300	[MPa]
12.4	Tensile Strength, Yield	Rp0.2	150	150	[MPa]
12.5	Poison's Ratio		0.25	0.25	
12.6	Contact Fatigue Limit	SHlim	360	360	[MPa]
12.7	Bending Fatigue Limit	SFlim	120	120	[MPa]
12.8	Tooth Hardness - Side	VHV	240	240	[HV]
12.9	Tooth Hardness - Core	JHV	240	240	[HV]
12.10	Base Number of Load Cycles in Contact	NHlim	5.00E+07	5.00E+07	
12.11	Wohler Curve Exponent for Contact	qH	10	10	
12.12	Base Number of Load Cycles in Bend	NFlim	3.00E+06	3.00E+06	
12.13	Wohler Curve Exponent for Bend	qF	6	6	

Additions section

13.0 ☑ Power, warming-up, gearbox surface

13.1	Ambient air temperature	20.00		[°C]
13.2	Maximum oil temperature	60.00		[°C]
13.3	Coefficient of heat dissipation	10.00		[W/m2/K]
13.4	Power losses	0.00		[kW]
13.5	Gearbox surface (min.)	0.00		[m2]

14.0 Preliminary design of shaft diameters (steel)

Recommended shaft diameter for:

14.1	- Main power-transmitting shafts	DA	23.43	23.36	[mm]

14.2	- Small, short shafts	DB	18.21	18.15	[mm]	

15.0 ☑ **Auxiliary calculations**

15.1	Transmission ratio calculation using the number of teeth	z1,z2 = i	14	43	= 3.0714
15.2	Transmission ratio calculation using the speed	n1,n2 = i	2000.0	750.0	= 2.6667
15.3	Power calculation using the pinion speed and torque moment	Mk1,n1=Pw1	270.0	1600.0	= 45.2356

16.0 ☑ **Graphical output, CAD systems**

16.1	2D drawing output to:	DXF File
16.2	2D Drawing scale	Automatic
16.3	Detail:	Gear

